Standard form of pair of linear equations

$$a_1x + b_1y + c_1 = 0$$
$$a_2x + b_2y + c_2 = 0$$

Here, a_1, b_1, a_2 and b_2 are non-zero real numbers

Experiment No. 5

Dated: 20 August 2025

Objective: To obtain the conditions for number of solutions / Inconsistency of a pair of linear equation in two variables by an activity.

Test activity

Conditions for Consistency / Inconsistency of Pair of Linear Equations

- Pre-Requisite Knowledge (with theory):

i) Coordinate system

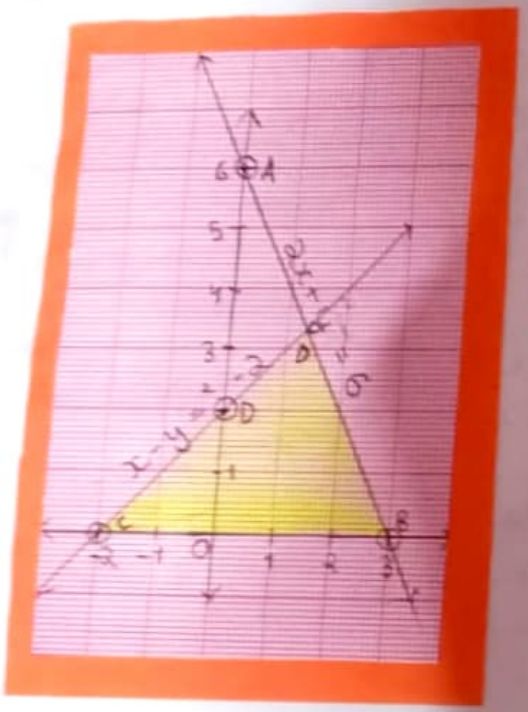
ii) Concept of plotting the points on a graph paper.

iii) An equation of the form $ax + by + c = 0$ is called linear equation in two variables, where a, b and c are real numbers and $a, b \neq 0$

iv) The graph of a linear equation in two variables is always a straight line.

- Materials Required:

Graph Paper, Geometry box, scale, paper, ~~scr~~ pen/pencil/stick



Objective :

Dated :

- Pair of linear equations : (Pair 1)

i) $2x + y = 6$

ii) $x - y = -2$

$\frac{a_1}{a_2} = \frac{2}{1}$; $\frac{b_1}{b_2} = \frac{1}{-1}$; $\frac{c_1}{c_2} = \frac{-6}{2} = \frac{-3}{1}$

$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$: Observation

Result : lines are intersecting

Number of zeroes : 1

→ Application : Area of between lines and x-axis

→ $\triangle ABC$ formed

Area $= \frac{1}{2} bh = \frac{1}{2} \times 5 \times 3.3$

$= \boxed{8.25 \text{ cm}^2}$

- Pair of linear equations : (Pair 2)

i) $2x + 4y - 4 = 0$

ii) $3x + 6y - 12 = 0$

$$\frac{a_1}{a_2} = \frac{1}{3} ; \frac{b_1}{b_2} = \frac{4}{6} = \frac{2}{3} ; \frac{c_1}{c_2} = \frac{-4}{-12} = \frac{2}{6} = \frac{1}{3}$$

Observation : $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

Result : 0 lines are parallel

Number of zeroes : 0

→ Application : Area between lines and axes

→ Solution :

Area of Trapezium ABCD = ar(ΔDOB) - ar(ΔCOA)

$$= \frac{1}{2} bh - \frac{1}{2} bh$$

$$= \frac{1}{\cancel{2}} \times 4 \times \cancel{2} - \frac{1}{\cancel{2}} \times \cancel{2} \times 1$$

$$= 4 - 1$$

$$= \boxed{3 \text{ cm}^2}$$

Experiment No. 5

Objective :

Dated :

- Pair of linear equations: (Pair 3)

i) $2x + 4y - 4 = 0$

ii) $4x + 8y - 8 = 0$

$\frac{a_1}{a_2} = \frac{2}{4} = \frac{1}{2}$; $\frac{b_1}{b_2} = \frac{4}{8} = \frac{1}{2}$; $\frac{c_1}{c_2} = \frac{\cancel{-1}}{\cancel{-2}} = \frac{1}{2}$

Observation: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

Result: Co-inciding lines

Number of zeroes: Infinitely many

→ Application: Area of triangle ΔABO formed between lines and axes

→ Solution:

$ar(ABO) = \frac{1}{2} bh$

$= \frac{1}{\cancel{2}} \times \cancel{2} \times 1 = \boxed{1 cm^2}$